

Of my $k = 5$ real facilities, how many should I place near each of the 5 rough centers `A, B, C, D, E`?"

A partition is one possible answer to that question. It is just a tuple `(k\_A, k\_B, k\_C, k\_D, k\_E)` of non-negative integers whose sum is k :

Line 5 of Algorithm 1 quantifies over all ordered tuples `(k\_1, k\_2, ..., k\_τ)` of non-negative integers summing to `k`. By the classical "stars and bars" identity, the number of such tuples is

...

$$k_A + k_B + k_C + k_D + k_E = 5$$

...

Examples for our problem ($k = 5, \tau = 5$):

| partition | meaning |

| `(1, 1, 1, 1, 1)` | one facility per rough center |

| `(5, 0, 0, 0, 0)` | put all 5 near A, none anywhere else |

| `(2, 1, 1, 1, 0)` | 2 near A, 1 each near B, C, D, none near E |

| `(0, 0, 5, 0, 0)` | all 5 near C |

| `(3, 1, 1, 0, 0)` | 3 near A, 1 each near B and C |

Each partition produces a *different* network of "where can a facility be opened?" and therefore a *different* possible assignment of cities to facilities, *different* cluster radii, and *different* capacity feasibility. The algorithm doesn't know up front which partition gives the smallest cost, so line 5 says **try them all** and line 17 says **keep the best**.

How many such partitions are there? the formula $C(2k-1, k-1)$

5 for each $(k_1, k_2, \dots, k_\tau)$ such that $\sum_{i=1}^\tau k_i = k$ do

stars and bars

Imagine you have:

* k indistinguishable **stars** (the k facilities you have to place), and
 * $\tau - 1$ indistinguishable **bars** (dividers that separate the τ rough centers from each other).

...

★ ★ ★ ★ ★ ← 5 stars (facilities)
 |||| ← 4 bars (dividers between A,B,C,D,E)
 ...

Some example arrangements:

...

★ ★ ★ ★ ★	→ (1, 1, 1, 1, 1)	balanced
★ ★ ★ ★ ★	→ (5, 0, 0, 0, 0)	all in A
★ ★ ★ ★ ★	→ (2, 1, 1, 1, 0)	
★ ★ ★ ★ ★	→ (0, 0, 5, 0, 0)	all in C
★ ★ ★ ★ ★	→ (3, 1, 1, 0, 0)	
...		

The total length of the row is $k + \tau - 1$ symbols ($5 + 4 = 9$ in our example). Out of these 9 positions we just have to **choose where the $\tau - 1$ bars sit**. So the count is:

> ## Number of partitions = $C(k + \tau - 1, \tau - 1)$

When $\tau = k$ (our case) this simplifies to:

> ## $C(2k - 1, k - 1)$

5.2 Plugging in numbers

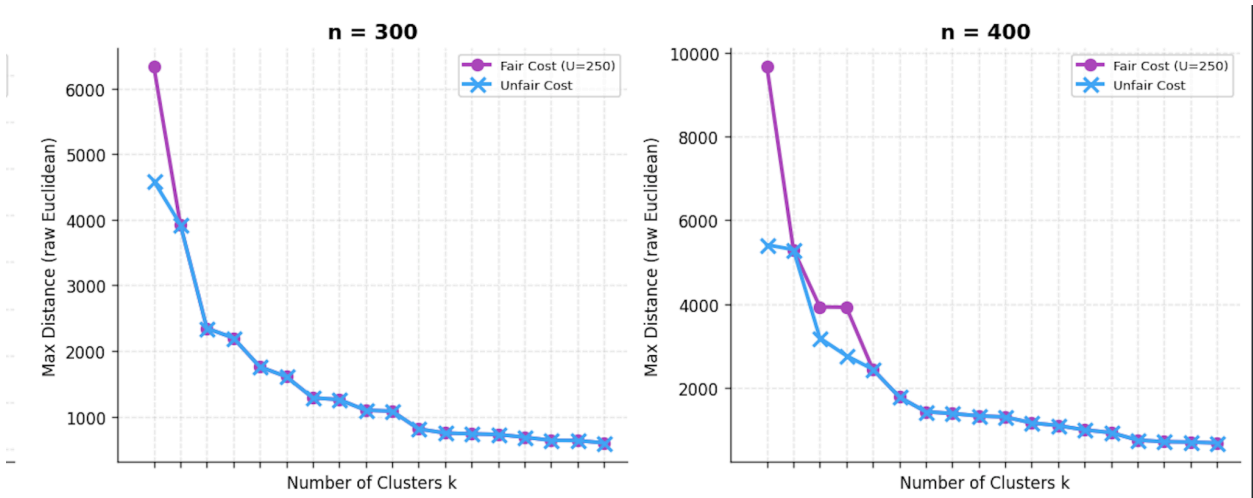
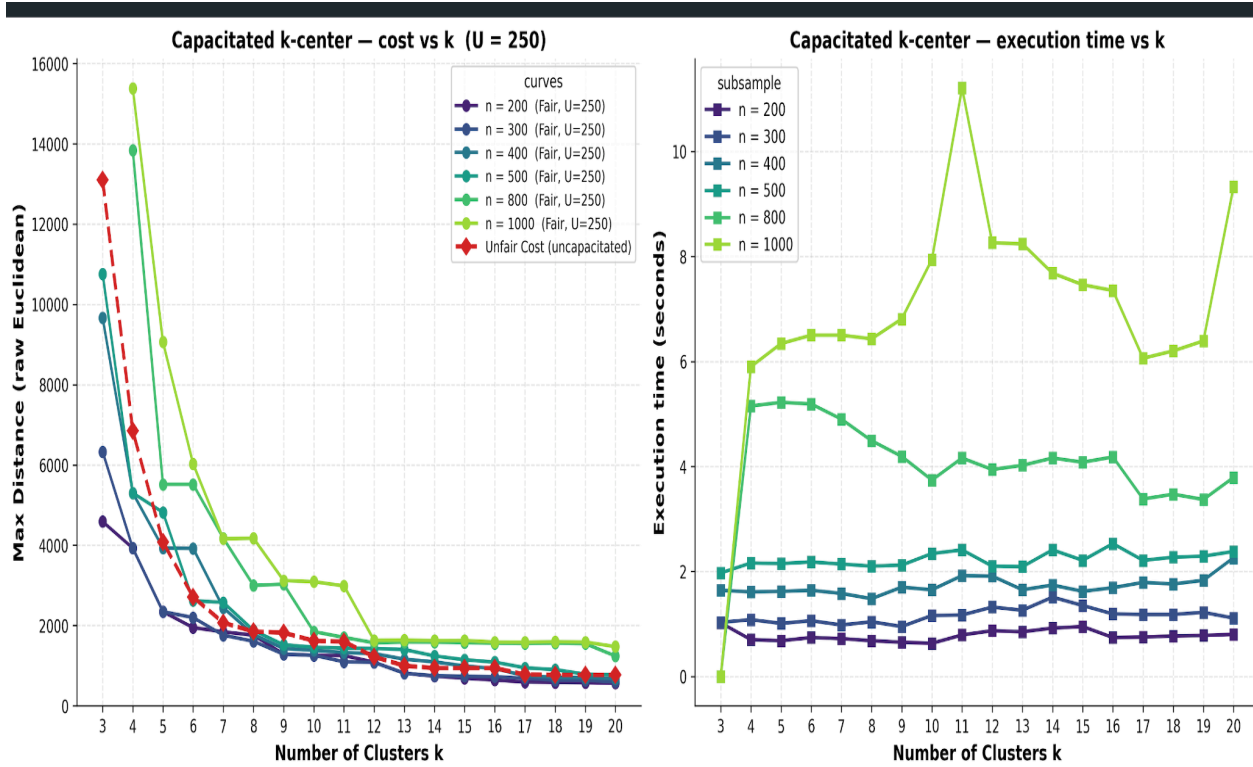
k partition row	total partitions
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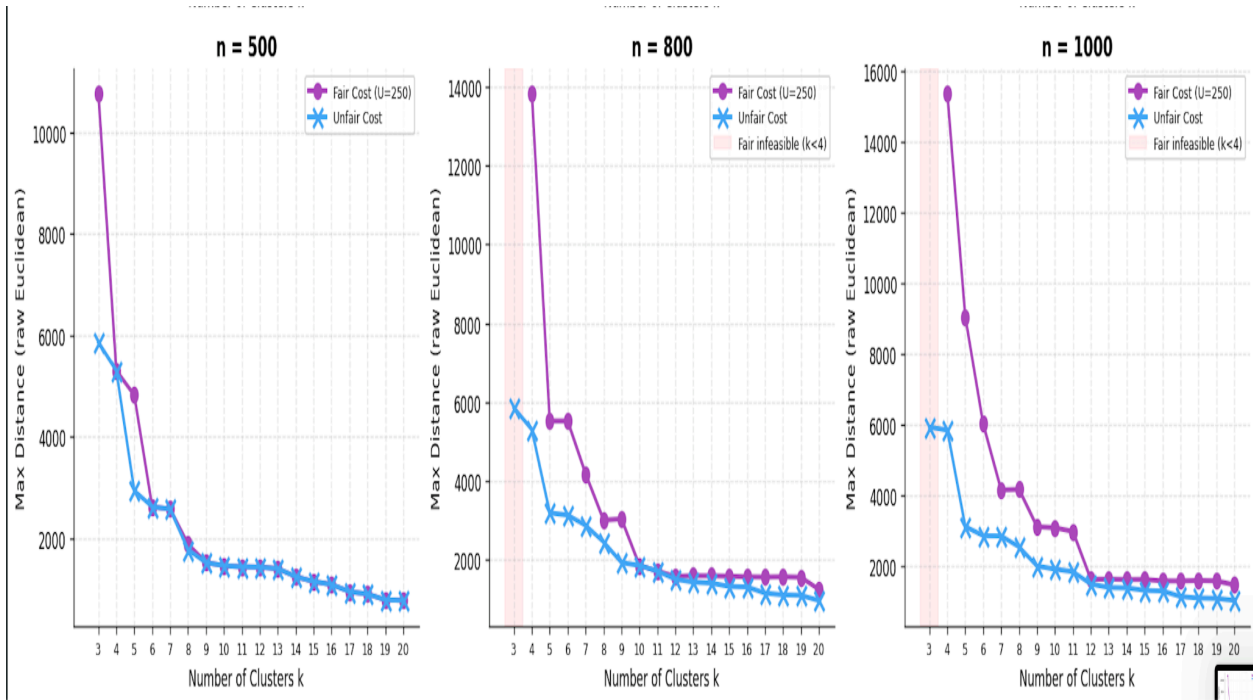
|---|---|---:|

`k` (= τ)	Number of partitions to enumerate

| 3 | 3 stars + 2 bars = 5 symbols, choose 2 | ` $C(5, 2) = 10` |
| 4 | 4 + 3 = 7, choose 3 | ` $C(7, 3) = 35` |$$

k	Count	
3	10	
4	35	
5	126	
8	6,435	
10	92,378	
12	1,352,078	
14	20,058,300	≈ 20.06 million partitions
16	300,540,195	≈ 300.54 million partitions
18	4,537,567,650	≈ 4.54 billion partitions
20	68,923,264,410	≈ 68.92 billion partitions





max_partitions = 8
max_r_samples = 60

